DP Biology – C2.1 (HL only)

Chemical signalling

- How can cells tell apart the many signals reaching them?
- What happens inside animal cells when a chemical signal arrives?

HL statements C2.1.1–14. Textbook pp. 450–466.



Image: Hawaiian bobtail squid, home of glowing Vibrio fischeri bacteria. Nick Hobgood, CC BY-SA 3.0.

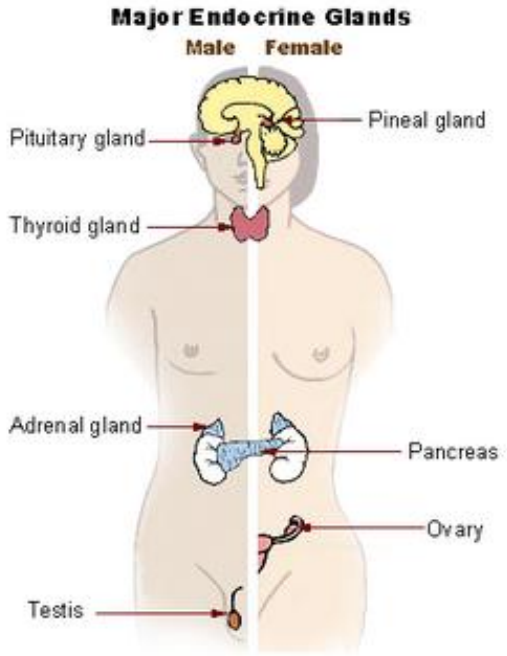
Objectives for each class

Class	Statements	By the end of the class, you can...
1	C2.1.1–5	Explain receptors and ligands, quorum sensing, the kinds of signalling chemical, and local versus distant signals
2	C2.1.6–9	Compare receptor types, and explain signal transduction, the acetylcholine receptor and G proteins
3	C2.1.10–14	Explain adrenaline, insulin and steroid hormone signalling, and feedback in signalling pathways

How cells talk

C2.1.1–14

Signal (ligand) released → Travels to target cell → **Binds a specific receptor** → Pathway inside the cell → Response



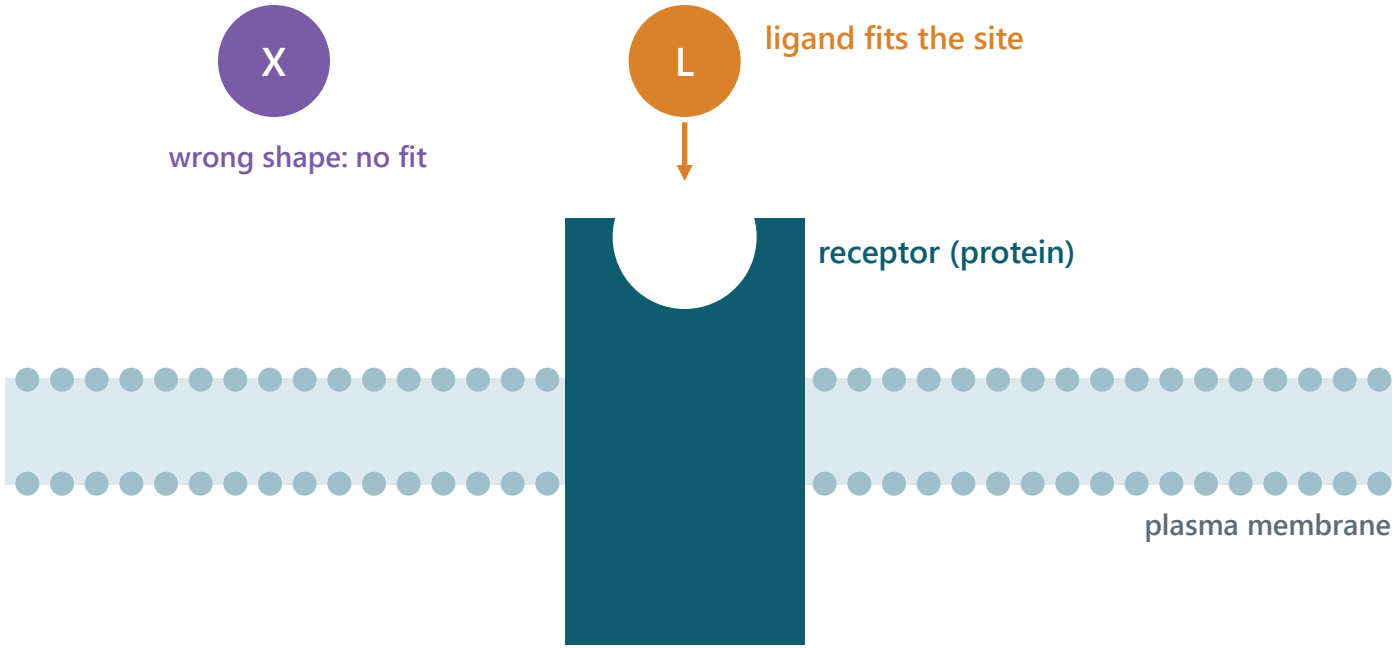
Endocrine glands. US Government, public domain.

Cells are bathed in many signals at once. Each cell only responds to the signals it has **receptors** for.

The signal usually does not enter the cell. Instead, binding to the receptor sets off a chain of events inside: **signal transduction**.

Receptors and ligands

C2.1.1



A **ligand** is a signalling chemical. A **receptor** is a protein with a binding site that fits only one ligand.

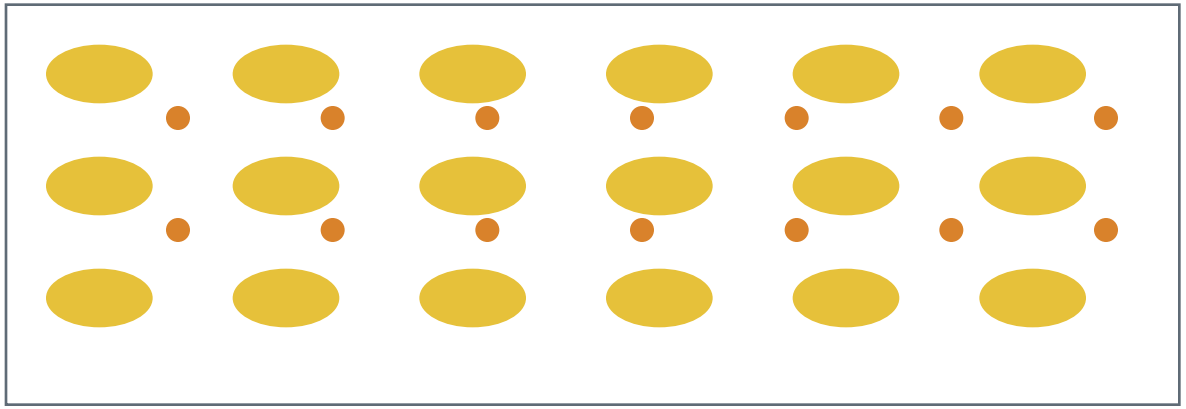
Binding changes the receptor's shape, which starts the response.

Unlike an enzyme, the receptor does not change the ligand.

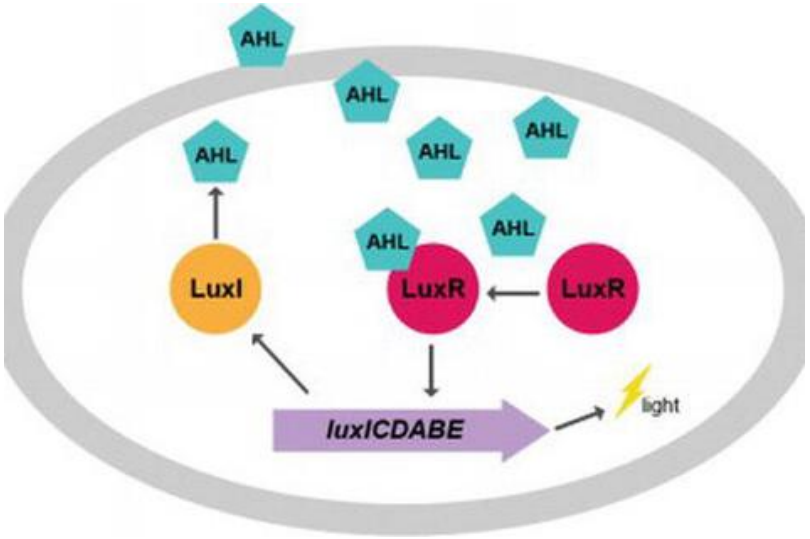
Quorum sensing: bacteria counting



few bacteria: signal dilute, no light



many bacteria: signal passes threshold, all glow



Quorum sensing genes. Wang, Payne and Bentley, CC BY 3.0.

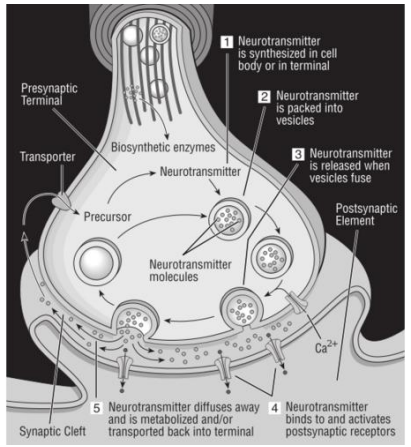
Each **Vibrio fischeri** cell releases a small signal molecule. When enough cells are packed together, the signal reaches a **threshold**, binds receptors, and switches on the light genes in all of them at once.

In the bobtail squid's light organ, the glow hides the squid's shadow from predators below.

Four kinds of signalling chemical

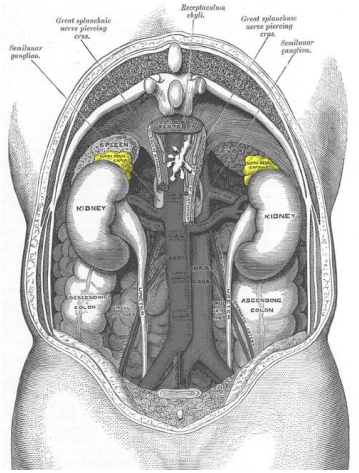
C2.1.3

Hormones	Neurotransmitters	Cytokines	Calcium ions
Glands → blood → distant targets	Neuron → across a synapse	Immune and other cells → nearby cells	Released inside a cell
e.g. insulin, adrenaline	**e.g. acetylcholine, dopamine**	**e.g. interleukins**	**e.g. trigger contraction, exocytosis**



Synapse. CC BY 4.0.

A neurotransmitter crosses a gap of about 20 nm. A hormone such as adrenaline is made in the adrenal glands and carried in the blood to the whole body.



Adrenal glands (yellow). Gray's Anatomy, public domain.

Many different chemicals

C2.1.4

Type	Examples
Amines	adrenaline, thyroxine, dopamine
Peptides and proteins	insulin, ADH, endorphins
Steroids	oestradiol, progesterone, testosterone
Amino acids	glutamate, GABA
Gas	nitric oxide

Why so many?

Each signal needs its own receptor shape. Chemistry also decides how a signal acts: steroids pass through membranes; proteins cannot; a gas spreads fast but only nearby.

Near and far

C2.1.5

	Neurotransmitter	Hormone
Distance	across one synapse	whole body, in blood
Speed	milliseconds	seconds to hours
How long it lasts	removed quickly	can last hours
Targets	one or a few cells	every cell with the receptor

Example

Acetylcholine crosses a 20 nm gap to one muscle fibre. Luteinizing hormone travels from the pituitary in the head to the ovaries or testes.

Working with data: quorum sensing

C2.1.2

Cell density and light output of Vibrio fischeri in a flask. These data are invented for practice; the pattern is realistic.

Time / h	Million cells per cm^3	Light
0	1	0
2	4	0
4	15	1
6	60	40
8	200	180
10	260	220

1. Describe the relationship between density and light.
2. Explain why there is almost no light for 4 hours.
3. Suggest the advantage of glowing only at high density.
4. Predict the effect of adding the signal at 0 h.

Check-in: class 1

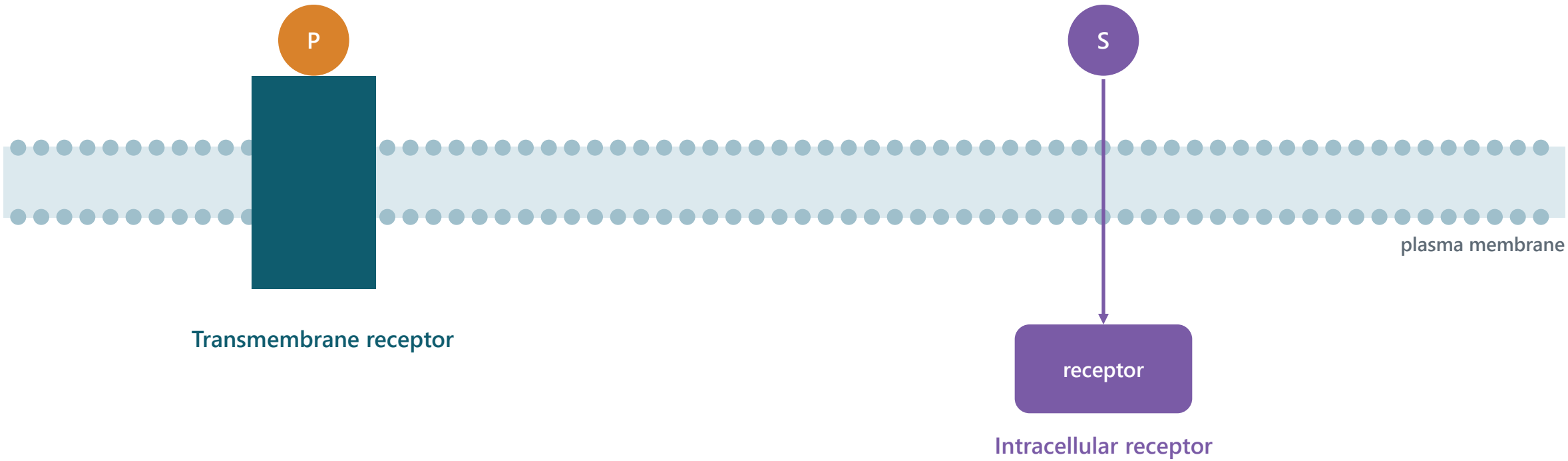
C2.1.1–5

1. Define ligand and receptor.
2. Explain how quorum sensing lets bacteria act together.
3. Name the four kinds of signalling chemical, with an example of each.
4. Explain why steroid hormones can enter cells but insulin cannot.
5. Compare neurotransmitters and hormones.

How to use this

Answer without your notes, then check against slides 4–9.

Receptors outside or inside?



	Transmembrane	Intracellular
Ligand	hydrophilic (P: protein)	hydrophobic (S: steroid)

Signal transduction

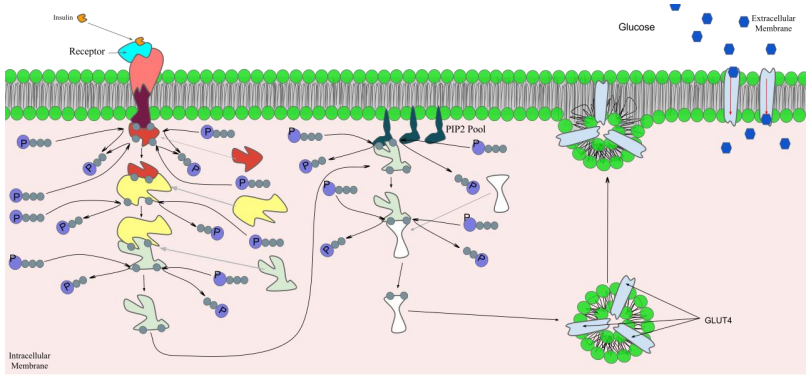
C2.1.7

Ligand binds → **Receptor changes shape** → Relay molecules activated inside → Signal amplified → Response

Signal transduction passes the message from the receptor into the cell, usually without the ligand entering.

Each activated molecule switches on many of the next, so a tiny signal produces a big response: **amplification**.

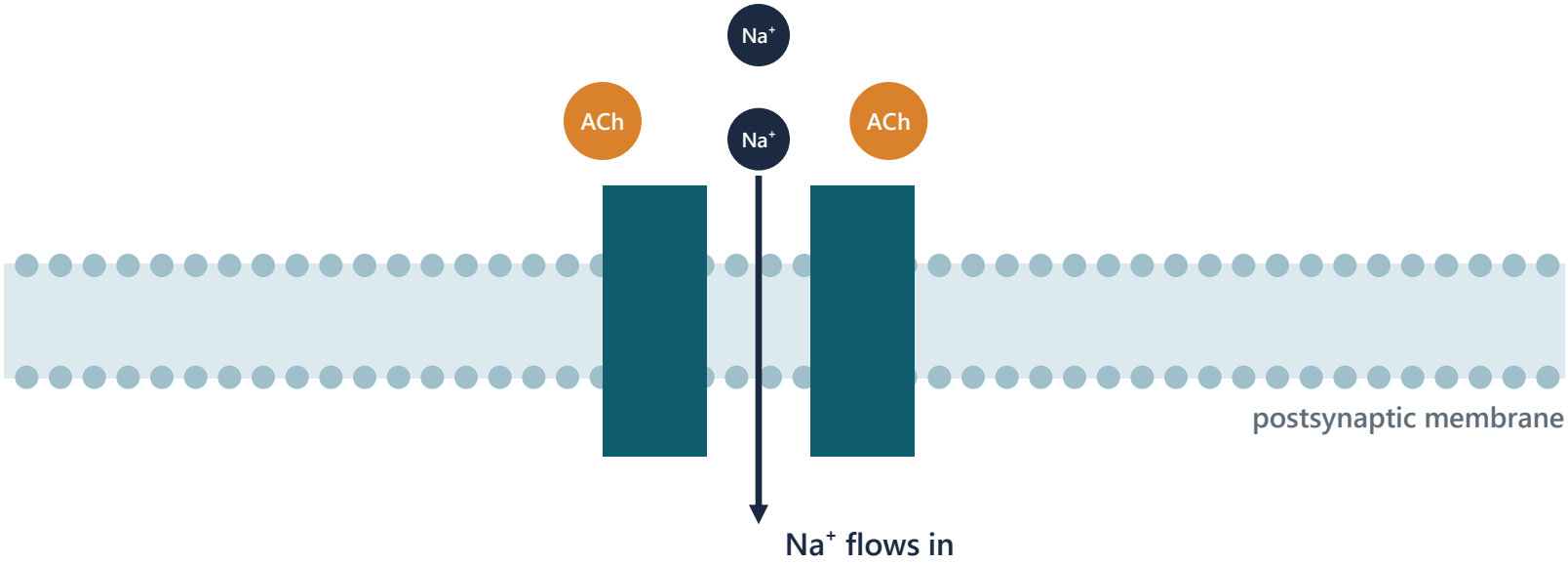
The response may be opening a channel, activating enzymes, or changing which genes are expressed.

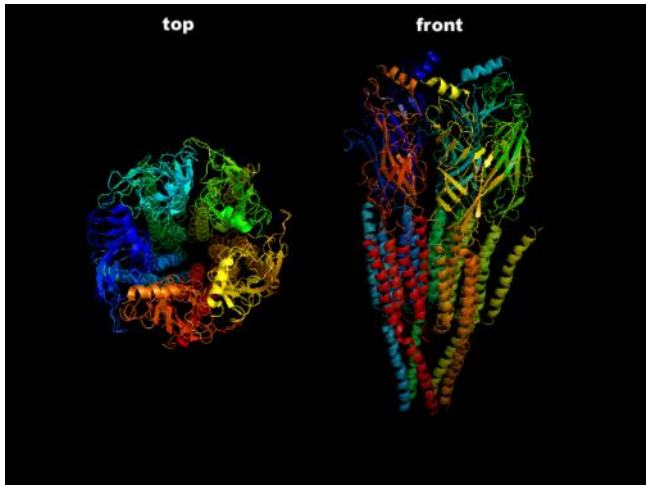


An example pathway (insulin). CC BY-SA 4.0.

The acetylcholine receptor: a gate

C2.1.8





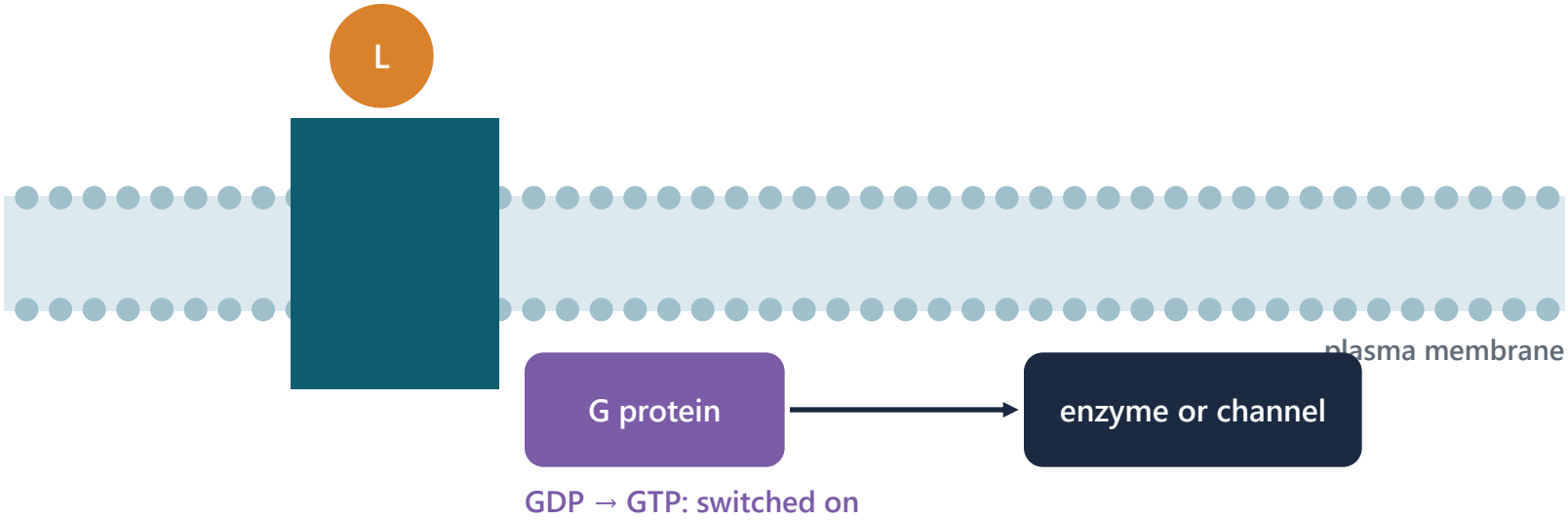
Acetylcholine receptor structure. Public domain.

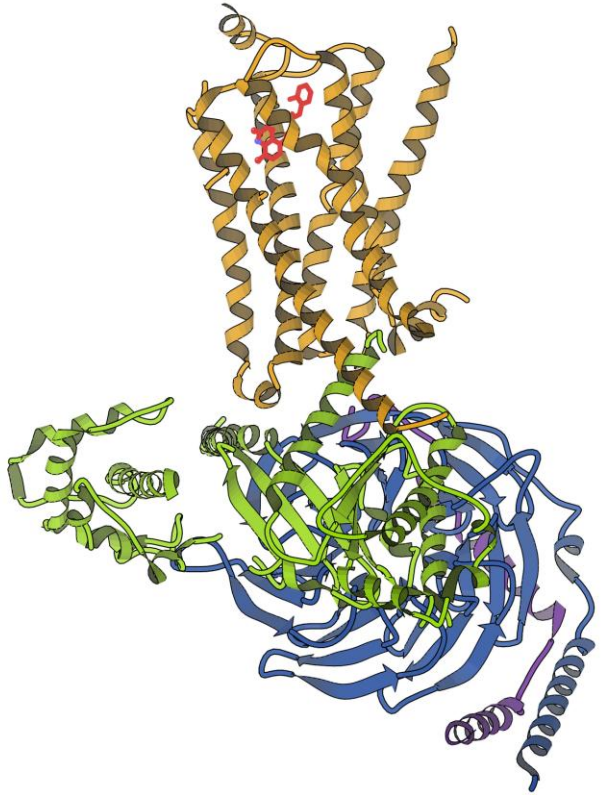
Acetylcholine binds the receptor, which **opens its own ion channel**. Sodium ions rush in and the inside becomes less negative (**depolarization**). If enough channels open, a nerve impulse or muscle contraction starts.

The receptor and channel are one protein, so this is very fast.

G protein-coupled receptors

C2.1.9





GPCR (gold) with its G protein (green, blue). CC BY 3.0.

The ligand binds outside; the receptor changes shape inside and switches on a **G protein** by swapping GDP for GTP. The G protein then activates an enzyme or channel.

Humans have about 800 of these receptors: for hormones, smells, tastes and light.

Practice: how big is amplification?

C2.1.7, 10

Approximate multiplication at each step after one adrenaline molecule binds (rounded values).

Step	Each activates
Receptor → G proteins	10
Adenylyl cyclase → cAMP	100
Protein kinase A → next enzymes	10
Phosphorylase → glucose released	1000

1. Calculate how many glucose molecules one adrenaline molecule could release.
2. Explain why amplification is useful in fight or flight.
3. Suggest why the signal must also be switched off quickly.

Check-in: class 2

C2.1.6–9

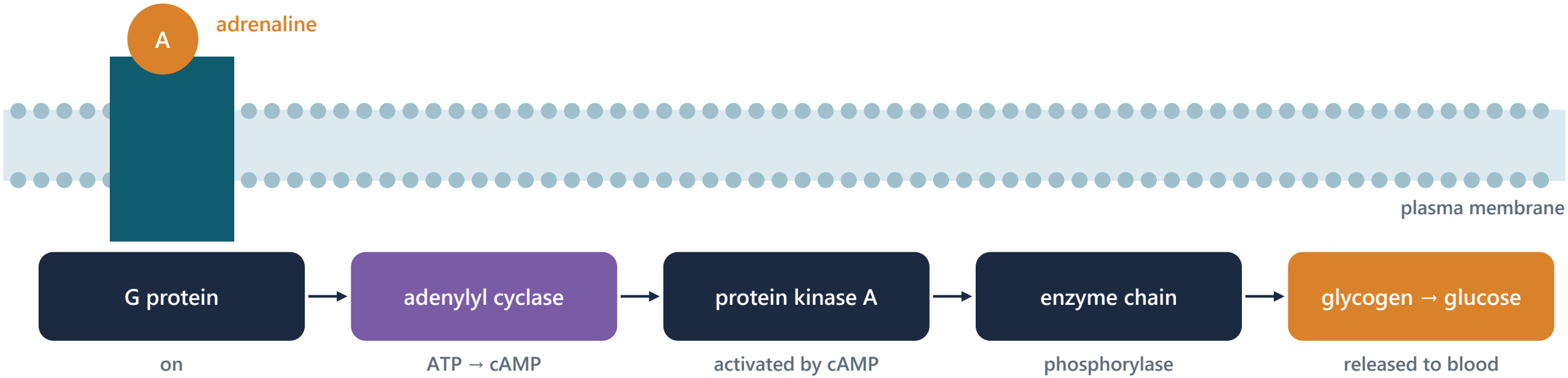
1. Distinguish between transmembrane and intracellular receptors.
2. Explain what is meant by signal transduction.
3. Describe how acetylcholine changes the membrane potential.
4. Outline how a G protein-coupled receptor passes on a signal.
5. Explain what is meant by amplification.

How to use this

Answer without your notes, then check against slides 11–15.

How adrenaline releases glucose

C2.1.10

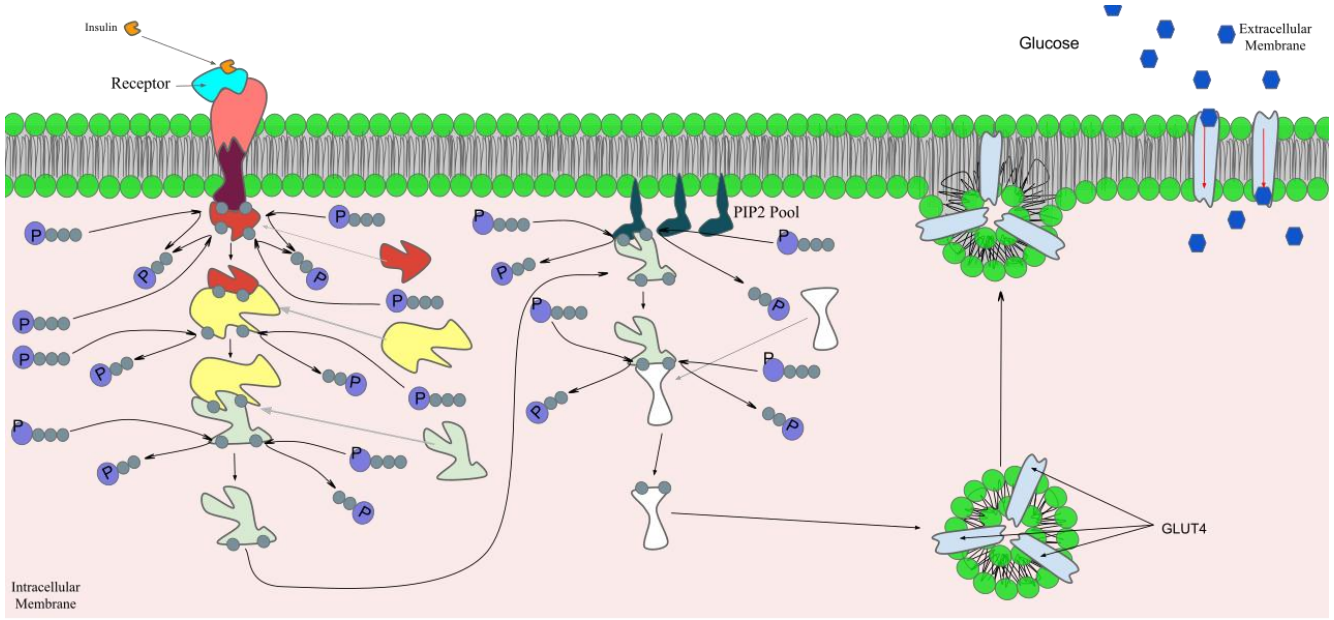


Adrenaline is the **first messenger**: it stays outside. Inside, **cyclic AMP (cAMP)** is the **second messenger** that carries the signal on.

In liver and muscle cells the result is fast breakdown of glycogen to glucose: fuel for fight or flight.

Insulin: a receptor that is an enzyme

C2.1.11



Insulin signalling. CC BY-SA 4.0.

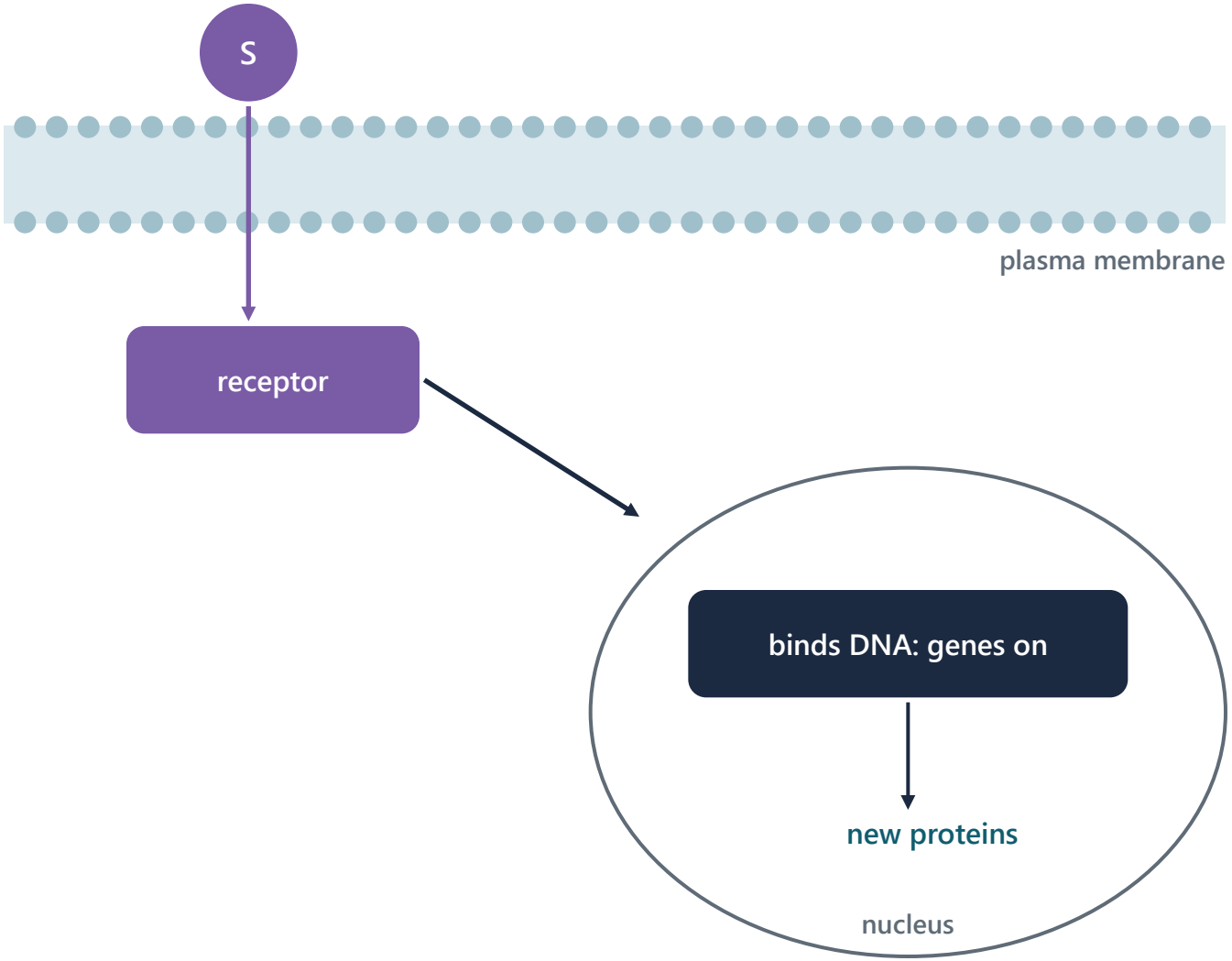
Compare

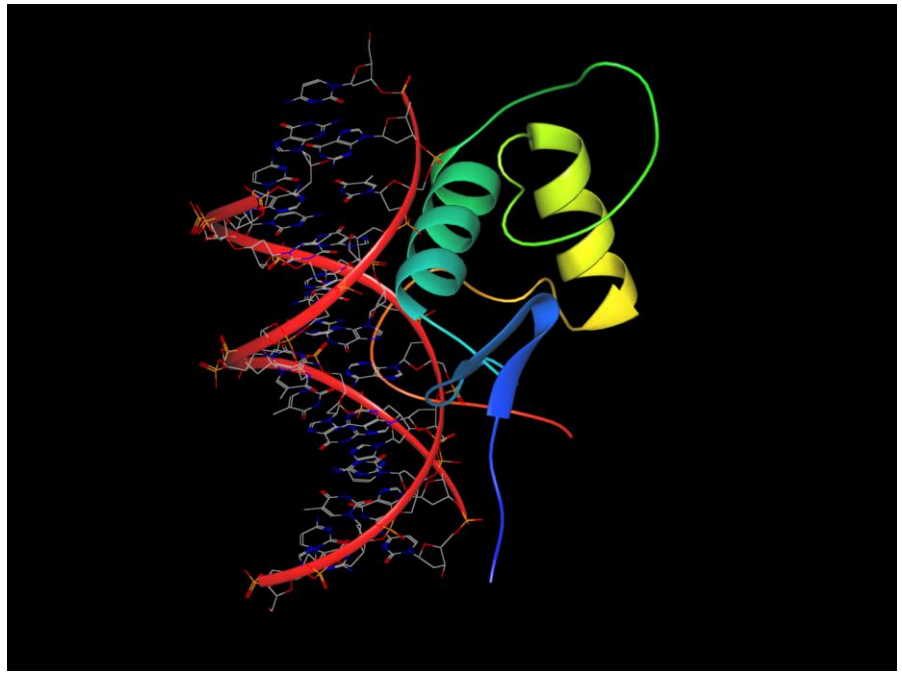
Adrenaline uses a G protein and cAMP. The insulin receptor is itself an enzyme: a **tyrosine kinase**. Result: muscle and fat cells take up glucose, lowering blood glucose.

Insulin binds → **Receptor adds phosphate to its own tyrosines** → Relay proteins start a cascade → GLUT4 vesicles fuse with membrane → More glucose enters

Steroid hormones switch genes on

C2.1.12





A steroid receptor bound to DNA. Public domain.

The steroid passes through the membrane, binds its receptor, and the pair acts as a **transcription factor**. Slower than other signals, but longer-lasting.

Oestradiol and progesterone

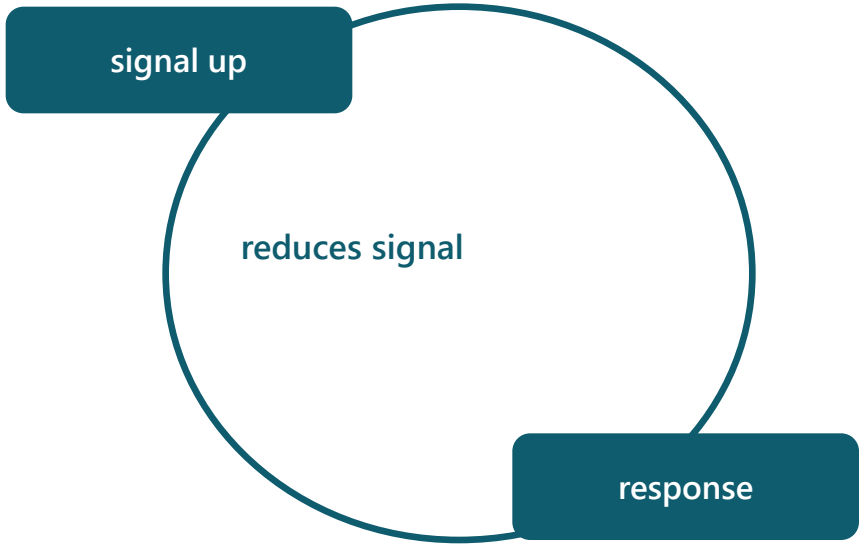
C2.1.13

	Oestradiol	Progesterone
Target cells	hypothalamus (GnRH-secreting cells)	endometrium (lining of the uterus)
Effect	changes GnRH release, which controls FSH and LH	keeps the lining thick, ready for an embryo

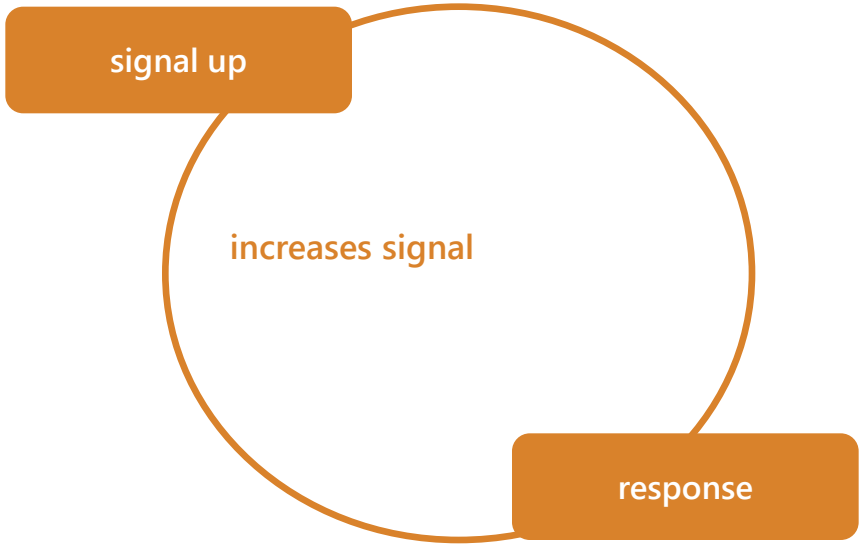
Same mechanism, different effects

Both bind intracellular receptors and change gene expression. The effect depends on the target cell: which receptors it has and which genes they switch on. This is part of the control of the menstrual cycle (D3.1).

Negative and positive feedback



Negative: keeps things stable



Positive: drives a rapid change

Example

Insulin lowers blood glucose, so less insulin is released.

Example

Quorum sensing signal makes more signal; oxytocin in childbirth makes more contractions.

Working with data: insulin and muscle cells

C2.1.11

Glucose uptake by muscle cells at different insulin concentrations. These data are invented for practice; the pattern matches real studies.

Insulin / pmol dm^{-3}	Uptake / units
0	10
50	35
100	58
200	80
400	85

1. Describe the effect of insulin on glucose uptake.
2. Explain why uptake levels off.
3. Explain why uptake is not zero without insulin.
4. Predict uptake in cells with fewer insulin receptors.

Four ways a signal acts

C2.1.6–13

Receptor	Example	How it works	Speed
Ion channel	acetylcholine	opens a channel; ions flow	milliseconds
G protein-coupled	adrenaline	G protein → cAMP → enzymes	seconds
Tyrosine kinase	insulin	phosphorylates itself → cascade	minutes
Intracellular	oestradiol	acts as a transcription factor	hours

Common mistakes to avoid

Not this	But this
The hormone enters the cell and acts directly.	Most bind a receptor; a pathway inside produces the response.
All receptors are on the cell surface.	Steroid receptors are in the cytoplasm or nucleus.
cAMP is the first messenger.	The hormone is first; cAMP is the second messenger.
Positive feedback keeps conditions stable.	Negative feedback stabilizes; positive feedback amplifies.

Check that you can...

1. **Explain** how receptors bind specific ligands
2. **Outline** quorum sensing in bacteria
3. **Classify** signalling chemicals and **compare** local and distant signals
4. **Distinguish** transmembrane and intracellular receptors
5. **Explain** signal transduction through ion channels, G proteins and tyrosine kinases
6. **Explain** how steroid hormones change gene expression
7. **Distinguish** positive and negative feedback

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Distinguish between a transmembrane and an intracellular receptor.
2. Outline the role of cAMP in the action of adrenaline.
3. Give one example each of positive and negative feedback in signalling.

Image credits

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Quorum sensing in Vibrio fischeri.png – Sally Wang, Gregory F. Payne and William E. Bentley – CC BY 3.0

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